

Canadian Journal of Fisheries and Aquatic Sciences

Decision Letter (cjfas-2026-0089)

From: cjfas@cdnsiencepub.com

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Subject: Decision on cjfas-2026-0089 - Canadian Journal of Fisheries and Aquatic Sciences

Body: 06-Jul-2026

Dear Mr. Ames,

cjfas-2026-0089: Employing deep learning for chum salmon (*Oncorhynchus keta*) age estimation through scale image analysis

We have now received advice on your manuscript from reviewers with expertise in the subject matter, and they have identified some significant concerns. However, if you feel that you can adequately respond to these concerns, we are prepared to consider a revised manuscript from you. In preparing your revision, please pay careful attention to the reviewers' comments, being guided by the Associate Editor's advice. The comments of the reviewers and the Associate Editor are included at the foot of this letter.

To help expedite the processing of your manuscript, please upload both clean and marked versions of your revised manuscript and provide a detailed, point-by-point reply to each of the comments, noting the line number(s) where each change was made. Please either specify the changes you have made to your manuscript or provide a reasoned rebuttal to each point that you feel is not justified. You should upload this response under "View and Respond to Decision Letter".

Revisions to your manuscript should be completed by 04-Sep-2026, unless you advise us that it will take longer. A manuscript received after this time with no prior communication from you may be treated as a new submission. (To add this due date to your Outlook or Google calendar, please open the attachment and click Save.)

To submit your revised manuscript, please log in to <https://mc06.manuscriptcentral.com/cjfas-pubs> and enter your Author Centre, where you will find your manuscript title listed under "Manuscripts with Decisions". Under "Actions", click on "Create a Revision". Your manuscript number has been appended to denote a revision.

You may also click the link below to start the revision process (or continue the process if you have already started your revision) for your manuscript. If you use the below link you will not be required to log in to ScholarOne Manuscripts.

*** PLEASE NOTE: This is a two-step process. After clicking on the link, you will be directed to a webpage to confirm. ***

https://mc06.manuscriptcentral.com/cjfas-pubs?URL_MASK=b6a4136043ad47beb4da6c3dedcb2764

IMPORTANT:

- Your original files are available to you when you upload your revised manuscript. Please delete any redundant files before completing the submission.

- If you do not plan to revise and resubmit this paper to Canadian Journal of Fisheries and Aquatic Sciences, please inform the editorial office (cjfas@cdnsiencepub.com) as soon as possible so that we can withdraw the paper.

Sincerely,
Martin Krkošek
Editor-in-Chief
Canadian Journal of Fisheries and Aquatic Sciences

NOTE: Due to file size limitations for email attachments, please visit your Author Dashboard to ensure that you have access to all files uploaded by editors and reviewers during the peer review of your manuscript. To access files, click "Manuscripts With Decisions," locate the manuscript, and click

"view decision letter." If applicable, files are attached to this email.

Associate Editor comments to author:

We now have two reviews of this manuscript. I appreciate the author's patience as we secured these as it took much longer than is typically the case. Both reviewers noted the manuscript is on a new and highly relevant topic that is likely to be the direction salmon scale ageing is headed in the future. They found the manuscript to be well written, and the conclusions supported by the results. They also identified several areas that need to be addressed before the manuscript is considered further, in addition to having numerous comments and suggestions.

Reviewer 1:

Aging of fish is a critical input into most modern stock assessments and hence sustainable fisheries management. However, ageing fish structures can be very time consuming and dependent on highly specialized human "agers" resulting in increasing gaps between information needs to inform management and what can be practically provided by traditional ageing techniques. This manuscript details the assembly of a large multi-agency dataset of aged chum scale samples, and evaluation of ability of machine learning models to correctly age them. The authors find that they have high agreement with expert-derived ages for dominant age classes and argue the models can help augment traditional ageing workflows and improve throughput.

I found the manuscript to be reasonably well written, the conclusions generally sound, and it's findings of likely broad interest to reader of CJFAS. I offer some major and minor suggestions to strengthen the manuscript below.

Major comments:

1. The authors argue the purpose of the manuscript was to "demonstrate value [of the CNN models] for increasing the throughput of salmon scale age estimation". However, they provide no direct quantification of this. The manuscript would be strengthened by describing how much faster (could be a quantitative or qualitative argument) the throughput would be with the CNN models that were developed. This would go a long way to convincing reader that throughput is indeed increased if the CNN models were adopted into modern agency led ageing workflows.
2. The central evaluation metric is agreement with human labels, not true age. It was a bit concerning that 29% of those scales with discrepancies in ages were deemed to be due to the original expert-derived ages not being supported by the subsequent expert-based reviews. To me this suggests that up to a third of the ages in the dataset may be incorrect/not supported by reviewers. The authors acknowledge that true ages are unknown in this context, but I encourage them to consider expanding the discussion of implications of learning systematic reader bias, and to highlight more clearly how disagreement rates compare to human-human precision.
3. I found the CNN background section overly detailed, and that there was some redundancy in deep learning explanations. I encourage authors to review the manuscript with a eye towards streamlining the methodological descriptions and to consider moving some text to the supplement.

Minor comments

- L 42: consider revising to: "Estimates of fish age are often critical data for stock assessments and hence central to sustainable fisheries management."
- L49: consider revising to: "...increasing demand for fish ages to support stock assessments..."
- L56: stating management of salmon relies of age estimates for in-season harvest management and recruitment estimates sells its uses short. I would revise the opening sentence to "Pacific salmon stock assessment and management typically relies on annual estimates of age from a large number of individuals".
- L59: define "escapement" for general audience at first use
- L74: "aged by human reader" instead of "labelled"? and then could delete subsequent sentence?
- L 81: what are "CNN layers"? consider defining for general audience
- L103: define "epoch" for reader
- L130-168: much of the details of the scale datasets could be moved to supplementary information and briefly described in the main text (along with table 1)
- L176-178: details of subsampling scale dataset for model development should go in next section on model development
- L195-198: describe for reader what "training" in this case actually means
- L224: which "model"s? The "complex architectures" described in previous sentence?
- L225: what is the "ImageNet dataset"?
- Methods: a schematic of the process used might be really helpful
- L264 "leading" instead of "leads"?
- L266: would "lowest" be less ambiguous than "best"?

Reviewer comments to author:

Reviewer: 1

Comments to the Author
See attached document.

Date Sent: 06-Jul-2026

File 1: [Author Due Date Canadian Journal of Fisheries and Aquatic Sciences.ics](#)

File 2: [- CJFAS Review.pdf](#)

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